Phosphate mineralisation

Breakdown of organic phosphates to form soluble inorganic phosphates is called mineralisation. Organisms produce phosphatase enzymes and catalyse mineralisation. The mineralised phosphates can be utilized by plants (Figure 11.5).

$$\text{Phytic acid} \xrightarrow[\text{Phytase}]{} \text{Inositol}$$

11.1.4 Sulphur Cycle

Sulphur is present in sulphur containing aminoacids. The sulphur cycle involves oxidation – reduction reaction between Sulphate (SO_4), Elemental S and H_2S and hence there is change in the valence states of sulphur from -2 to +6.

The basic steps involved in sulphur cycle are

1. Sulphide/ sulphur oxidation
2. Sulphate reduction
3. Sulphur reduction
4. Organic sulphur compound oxidation or reduction
5. Desulfurylation

Sulphide/Sulphur oxidation ($H_2S \rightarrow S^0 \rightarrow SO_4^{2-}$)

It is carried out by prokaryotes under aerobic and anaerobic conditions. Under aerobic conditions, H_2S is spontaneously oxidized at neutral pH to elemental sulphur. Elemental sulphur is oxidized to suphates by chemolithotrophic bacteria like *Thiobacillus, Beggiatoa.*

If light is available, H_2S can be used as electron donor to carry out photosynthesis

HOTS

If there is no biogeochemical cycle in the ecosystem, What will happen to the earth?

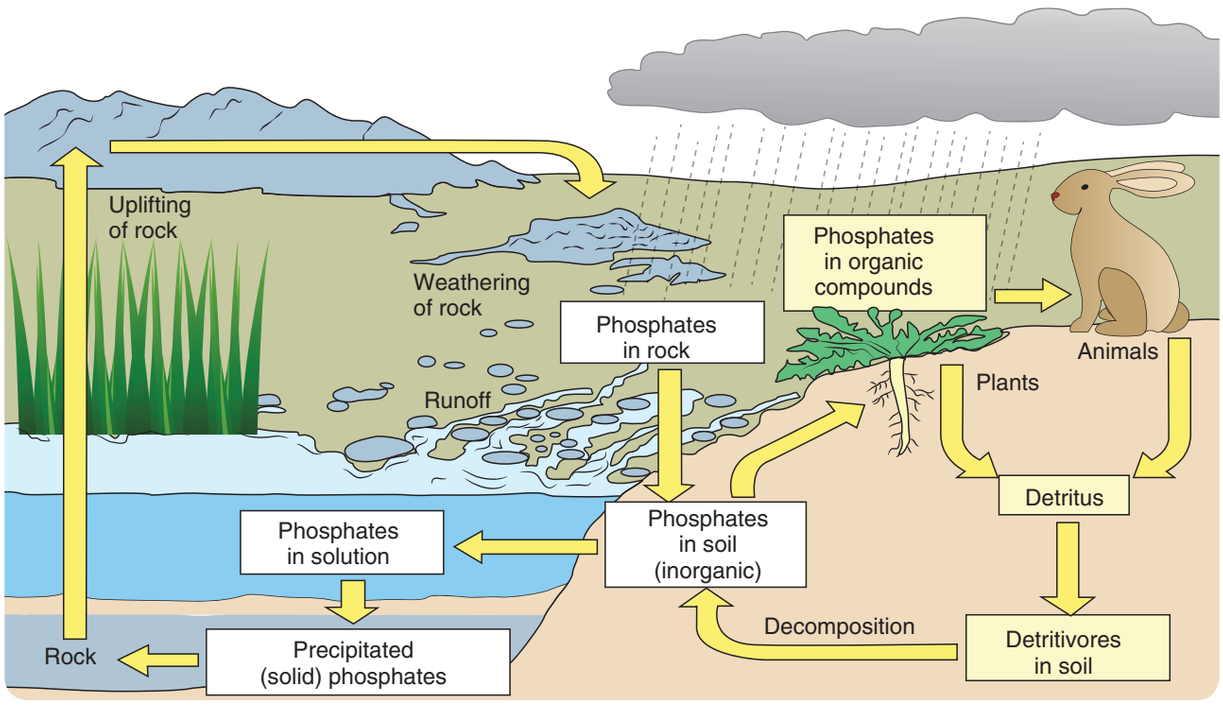


Figure 11.5: Phosphorus cycle

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